

Fast Maximum Likelihood Estimation for 1D Gaussian Splits

Nicola Greggio • Alexandre Bernardino

Institute for Systems and Robotics, Instituto Superior Técnico, Universidade de Lisboa

We propose GPM1D, an efficient method to estimate the parameters of a 1D mixture of two Gaussians. It can be used to replace the usual iterative EM method with an easily parallelizable and deterministic global maximum-likelihood search over feasible two-Gaussian pairs.

Why are mixtures of two Gaussians interesting ?

Incremental GMM learning starts coarse and increases model order one split at a time. Each split converts one Gaussian component in two new Gaussian children: if the two children follow the local data structure, global EM needs less correction and is less likely to settle in a poor basin.

What a good split must achieve

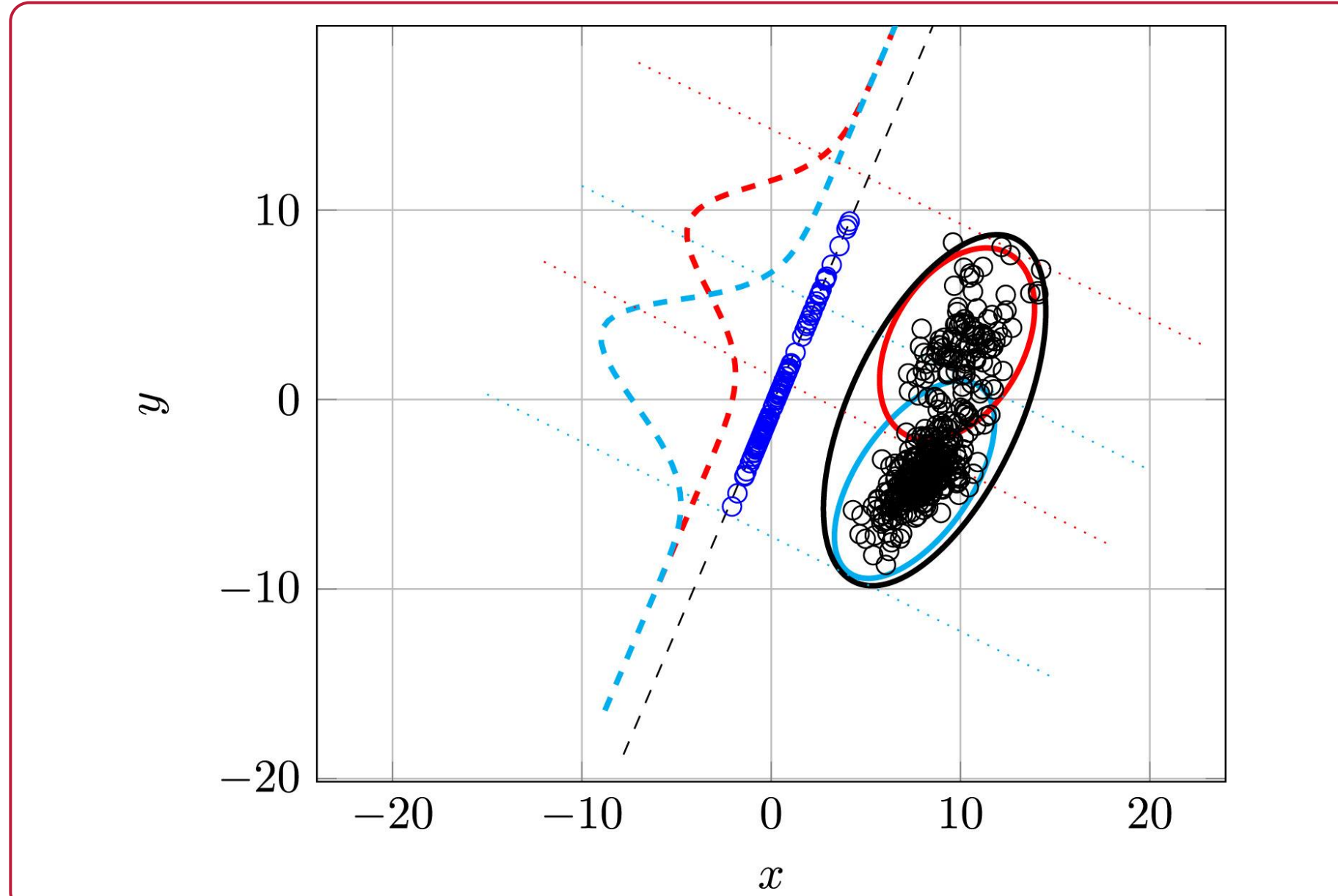
Create two valid children that preserve the parent statistics, explain its samples better, and begin close enough to the best fit that downstream EM only needs refinement.

Which problem does any split face

The infinity of solutions due to the ill-posedness of the split problem, that may lead to non optimal solutions.

Why are 1D splits interesting ?

A multivariate split is ill-posed and parameter-heavy. Projection onto a covariance eigenvector reduces it to a scalar fit; the principal direction is a strong first choice because it captures maximum variance and can expose hidden modes.



1D split move: project the parent component along a principal direction, fit two 1D Gaussians, then lift the children back to the original space. The search becomes much smaller while the orthogonal geometry is preserved. Adapted from Greggio & Bernardino (2024).

Main contributions

- Formulate the split as maximum-likelihood fitting of a normalized 1D two-Gaussian mixture.
- Use the parent moments to enumerate only feasible child parameters.
- Score every grid hypothesis independently and select the global optimum up to discretization.
- Provide a modular replacement for heuristic or EM-based 1D split initialization.

Method objective

PLACE THE CHILDREN NEAR THE BEST LOCAL FIT BEFORE GLOBAL EM.

REPRODUCIBLE • PARALLELIZABLE • PREDICTABLE
HYPOTHESIS BUDGET

How to use GPM1D

A generic greedy GMM algorithms selects a principal direction to split

Use a covariance eigenvector; the leading one captures maximum variance and is tested first.

Project and normalize

Project the samples belonging to that Gaussian component to 1D, along the identified principal direction, then set their mean to 0 and variance to 1.

GPM1D evaluates the best 1D Gaussians that fit these points:

Enumerate feasible children

Discretize child means and variances while enforcing the parent moment constraints.

Score each candidate

Compute the data log-likelihood for every feasible two-Gaussian hypothesis; candidates are independent.

Choose and lift back

Take the best grid hypothesis and map its two children back to the original multivariate space.

Gaussian-Pair mixture ML search

$$L(h) = \sum_i \log [w_1 \varphi(x_i; m_1, \sigma_1) + w_2 \varphi(x_i; m_2, \sigma_2)]$$

$$h^* = \arg \max_h L(h)$$

Predictable accuracy–runtime

For a chosen discretization of the mean and standard deviation (N_m , N_σ), the number of candidate grid points is known before execution. Unlike EM, there is no convergence-dependent loop. Finer grids approach the continuous optimum more closely; coarser grids trade precision for latency.

Example search sizes

$$\begin{aligned} N_m = 10, N_\sigma = 3 &\rightarrow 81 \text{ hypotheses} \\ N_m = 20, N_\sigma = 12 &\rightarrow 1,488 \text{ hypotheses} \\ N_m = 25, N_\sigma = 21 &\rightarrow 4,158 \text{ hypotheses} \end{aligned}$$

Example Application of GPM1D

Any greedy GMM description algorithm, such as:

- Greggio & Bernardino (2024): 1D EM gives a fast, robust two-Gaussian split initialization.
- GPM1D: feasible-grid scoring removes 1D EM initialization and convergence variability.
- Outcome: the projection stays efficient while the grid optimum is deterministic.

SAME 1D GEOMETRY
A STRONGER OPTIMIZER FOR THE SPLIT

Results

We benchmark the GPM1D itself against MATLAB's two-component EM on synthetic 1D mixtures. Isolating this step tests initialization quality before GPM1D is embedded in a full incremental estimator.

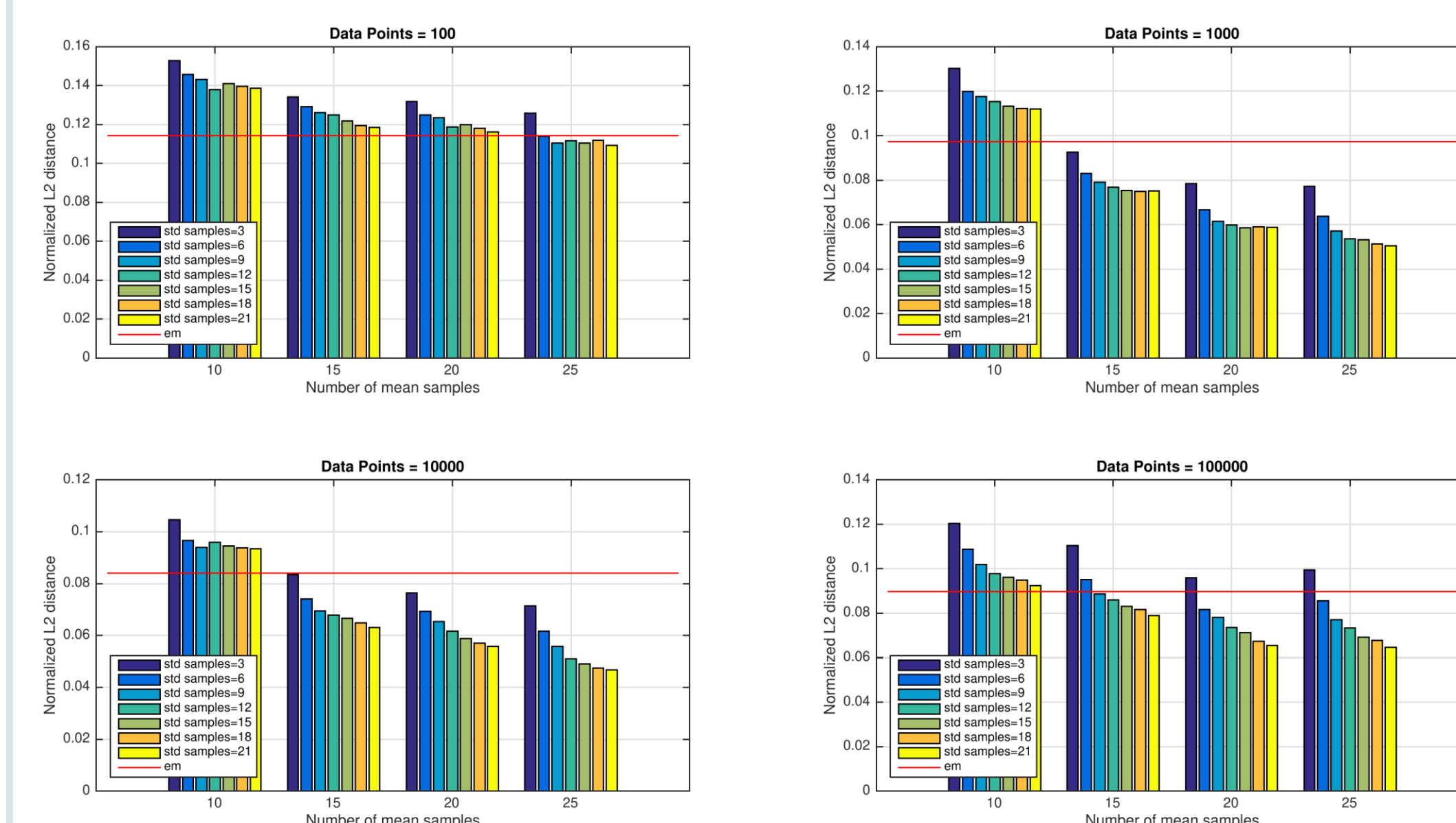
100 trials • 10^2 to 10^5 samples

$N_m \in \{10, 15, 20, 25\}$

$N_\sigma \in \{3, 6, 9, 12, 15, 18, 21\}$

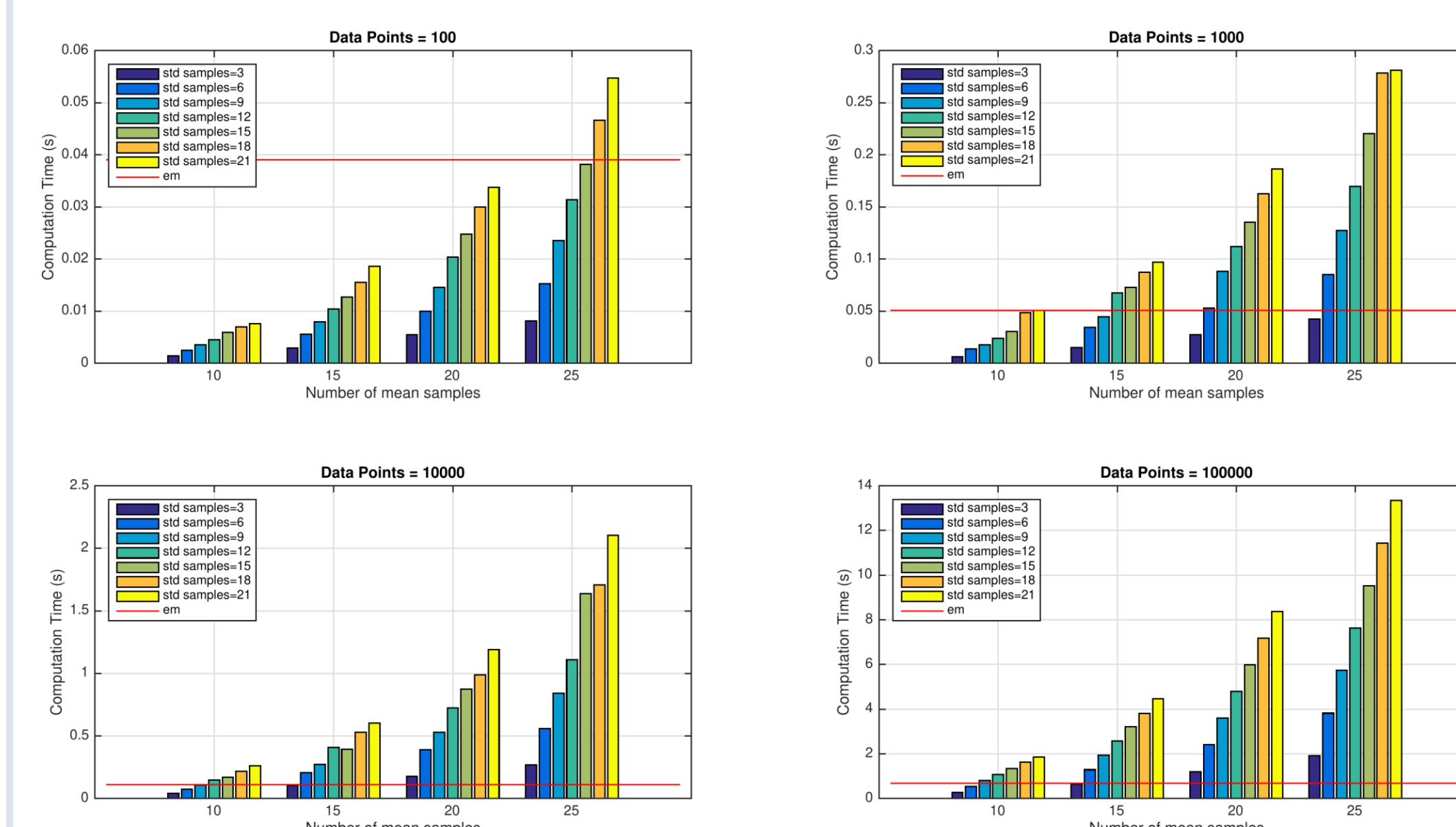
Metrics: normalized L2 distance, log-likelihood, runtime

Estimation accuracy



Finer grids lower normalized L2 error. From 1,000 samples onward, selected GPM1D settings beat EM, showing that deterministic grid search can initialize the split more accurately.

Computation time



At 100–1,000 samples, some grids are both faster and more accurate than EM. At 10,000–100,000 samples, the best accuracy costs more sequential time but the method can be easily parallelizable.

Impact of the results

- A better split gives global EM a closer starting point, reducing corrective re-estimation.
- A fixed grid makes runtime reproducible and tunable for latency-sensitive systems.
- A drop-in split can strengthen the 2024 FSAEM–EM pipeline and other greedy GMM learners.

Scope and next steps

Global optimality holds on the chosen grid, and one projection can miss multi-directional structure. Next: adaptive grids, parallel execution, and end-to-end tests in multivariate incremental GMM learning.